REVIEW

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How to Cure Autoimmune GN and Podocytopathies

Hans-Joachim Anders,[1] *Stefanie Steiger*,[1] *and Paola Romagnani*[2]

Abstract

Autoimmune GN and podocytopathies are immune-mediated kidney diseases with different clinical presentations and histotypes. Traditionally, proteinuria and histotypes are used for prognosis prediction and hence define intensity of immunotherapy. Renin-angiotensin system and sodium-glucose transporter 2 inhibitors are considered as "supportive care," and control of proteinuria seems a primary treatment goal without reasoning the cause of proteinuria. We propose to refine these concepts based on the shared pathophysiology of these diseases: (*1*) Disease acuity as the primary determinant of therapy. Rapidly progressive GN, relapsing GN, and chronic GN require different priorities. Rapidly progressive GN depends on the level and nephrotoxicity of the involved antibodies and complement activation and may require immediate complement inhibition, and antibody removal from the circulation before a B-cell–targeting therapy is initiated to control *de novo* autoantibody production. (*2*) Relapsing or chronically active disease need long-term control of immunologic activity with a B-cell–targeting monotherapy, in case of single autoreactive lymphocyte clones, for example, in antineutrophil cytoplasmic antibodies vasculitis or antinephrin/anti–M-type phospholipase A2 receptor-nephrotic syndrome. By contrast, diseases with numerous autoantigens/clones, *i.e.*, lupus nephritis or antiphospholipid syndrome should benefit from combination therapies, similar to kidney transplantation. (*3*) All forms of GN and most relapsing podocytopathies lead to glomerulosclerosis and nephron loss, *i.e.*, CKD. This implies CKD management following the latest Kidney Disease Improving Global Outcomes CKD risk matrix and treatment recommendations. In relapsing GN/podocytopathies, CKD care is the second treatment priority; in chronic GNs, it becomes the first treatment priority in contrast to "supportive care." In relapsing and chronic disease, proteinuria levels may represent activity, CKD, or both; hence, proteinuria alone does not inform treatment choices. This review aims to overcome existing hurdles by redefining treatment priorities in GNs and podocytopathies based on the underlying autoimmune pathomechanisms to define immunotherapy and by implementing CKD care for conceptual clarity and better long-term outcomes.

JASN 37: 394–404, 2026. doi: https://doi.org/10.1681/ASN.0000000912

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Introduction

GN is a group of inflammatory glomerular disorders characterized by the presence of proteinuria, indicating podocyte injury, and hematuria, reflecting injury to the glomerular basement membrane (GBM).[1] Leukocyturia is commonly present, too, a sign of glomerular inflammation and complement activation.[2,3] We distinguish five different categories of GN, each requiring different expert input and treatment approaches (Table 1)[1]: Infectious GN requires pathogen control to eliminate the source of the nephropathic pathogen components or immune complexes, maybe together with an infectious disease specialist.[4] Monoclonal GN requires a clone-directed therapy, together with a hematologist, to target the underlying plasma cell or B-cell clone that produces the nephropathic monoclonal Ig (component).[5] Alloimmune GN typically requires an immunosuppressive combination therapy and input from transplant experts, and genetic/autoinflammatory GN requires consultation of a geneticist and rare disease specialist.[6] Autoimmune GNs, which attract the most clinical attention, may best be managed together with an immunology expert, although nephrologists usually feel comfortable to manage them on their own (Table 2). Although this was acceptable at times when treatment was largely limited to steroids, cyclophosphamide, and inhibitors of the renin-angiotensin-aldosterone system, the expanding field of immunotherapeutics presents a growing challenge in selecting the most appropriate therapy.[7]

Autoimmune podocytopathies follow the same logic.[8,9] Although they may not present with proliferative glomerular lesions, nephrotic syndrome caused by autoantibodies against proteins of the slit diaphragm (*e.g.*, nephrin, podocin, and Kirrel1)[10,11] or the podocyte itself (*e.g.*, M-type phospholipase A2 receptor [PLA2R], THS7DA, neural epidermal growth factor–like 1 protein, and others)[12] involve the same type of immunopathogenesis and therefore respond to similar treatments (Figure 1).[13]

The growing understanding of the pathophysiology of autoimmune glomerular disorders and the rapidly

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Published Online Ahead of Print: October 10, 2025

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Table 1. Types of GN/podocytopathy and which specialist to involve

Type of GN/Podocytopathy	Supportive Specialist
Infectious GN/ podocytopathy	Infectiologist
Autoimmune GN/ podocytopathy	Immunologist
Alloimmune GN	Transplant specialist
Auto-inflammatory/genetic GN or podocytopathy	Geneticist
Monoclonal GN/ podocytopathy	Hematologist
Toxic podocytopathy	Toxicologist, pharmacist
Adaptive podocytopathy	Diabetologist or other depending from the cause of hyperfiltration

changing treatment landscape require a reconsideration of the traditional conceptual framework based on histotypes and proteinuria response. Achieving proteinuria remission alone is no longer adequate, if disease relapses occur or if eGFR continues to decline. Instead, the notion of a cure for autoimmune GNs/podocytopathies could become a realistic treatment goal.

What does "curing" GNs/podocytopathies mean? Well, a cure for autoimmunity would imply regaining tolerance and may require selectively depleting the antigen-specific lymphocytes clones[14,15] or at least part of the B-cell lineage, although the latter eliminates many homeostatic clones as well.[16] However, curing an autoimmune GN/podocytopathy may also mean minimizing irreversible kidney injury and loss of eGFR, preventing disease relapses, and maintaining a physiologic kidney lifespan, so that patients will not require KRT in the last years of life and experience cardiovascular events earlier compared with the healthy population.[17]

This review provides an update to the conceptual framework for the management of GN and podocytopathies with the goal of minimizing their short-term and long-term consequences. In contrast to Kidney Disease Improving Global Outcomes guidelines,[18] which rely on the historical histotype disease categories, we rather specifically discuss the category of the autoimmune GNs segregated by the level of immunologic activity versus the extent of irreversible nephron loss. The latter contributes to increased hemodynamic and metabolic workload on the remaining nephrons, driving an adaptive podocytopathy.[19] This approach helps to direct the respective management approach and to prioritize research questions to improve patient outcomes in autoimmune GN or podocytopathies.

Two Determinants of Acuity and Prognosis of Autoimmune Glomerular Disorders

The Nephrotoxicity of the Autoantibody

Whether an autoimmune GN/podocytopathy presents as rapidly progressive or chronic disease depends on the

Table 2. Types of autoimmune GN and podocytopathies

Clinical Presentation	Autoantigen (Compartment)	Serum Biomarker	Histotype
RPGN			
+ Lung disease	NC1-collagen 4, peroxidasin, perlecan (GBM)	Autoantibody	Crescentic GN with IgG along GBM
+ Vasculitis	Proteinase-3	ANCA, anti-PR3, anti-MPO	Pauci-immune crescentic GN
+ Vasculitis	Myeloperoxidase (blood neutrophils)		
+ Systemic disease	Chromatin components IgG (blood)	ANA, dsDNA, … cryoglobulins	Crescentic immune complex GN
+ Vasculitis	Complement regulators (blood)	Anti-C3/FH/FB/FD, …	Crescentic C3G
+ Thrombosis	Phospholipids (endothelium)	Anti PL-antibodies	Glomerular TMA
Relapsing disease			
Active GN	Proteinase-3 myeloperoxidase (blood neutrophils)	ANCA, specific autoantibody	Pauci-immune crescentic GN
Active GN	Chromatin components IgG (blood)	ANA, dsDNA, … cryoglobulins	Immune complex GN
Nephrotic syndrome	Nephrin, podocin, Kirrel1 (filtration slit)	Autoantibody	Minimal change "disease"
Nephrotic syndrome	PLA2R, NELL-1, THSD7A, SEMA3B, PCDH7, HTRA1, NTNG1 (podocyte)	Autoantibody	Membranous nephropathy
CKD			
	GdIgA1 (blood)	Anti-GdIgA1	IgA nephropathy
	Complement regulators (blood)	Anti-FH/FB/FD, …	C3G
	Nephrin, podocin, Kirrel1 (filtration slit)	Autoantibody	"Primary" FSGS

ANA, antinuclear antibodies; ANCA, antineutrophil cytoplasmic antibodies; anti-FH/FB/FD, autoantibodies against complement factor H, factor B, or factor D; C3G, C3 glomerulopathy; dsDNA, double-stranded DNA; GBM, glomerular basement membrane; GdIgA1, galactose-deficient Ig A1; HTRA1, high-temperature requirement A serine peptidase 1; MPO, myeloperoxidase; NELL-1, neural epidermal growth factor–like 1 protein; NTNG1, netrin G1; PCDH7, protocadherin 7; PLA2R, M-type phospholipase A2 receptor; PR3, proteinase 3; RPGN, rapidly progressive GN; SEMA3B, semaphorin 3B; THSD7A, thrombospondin type 1 domain–containing 7A; TMA, thrombotic microangiopathy.

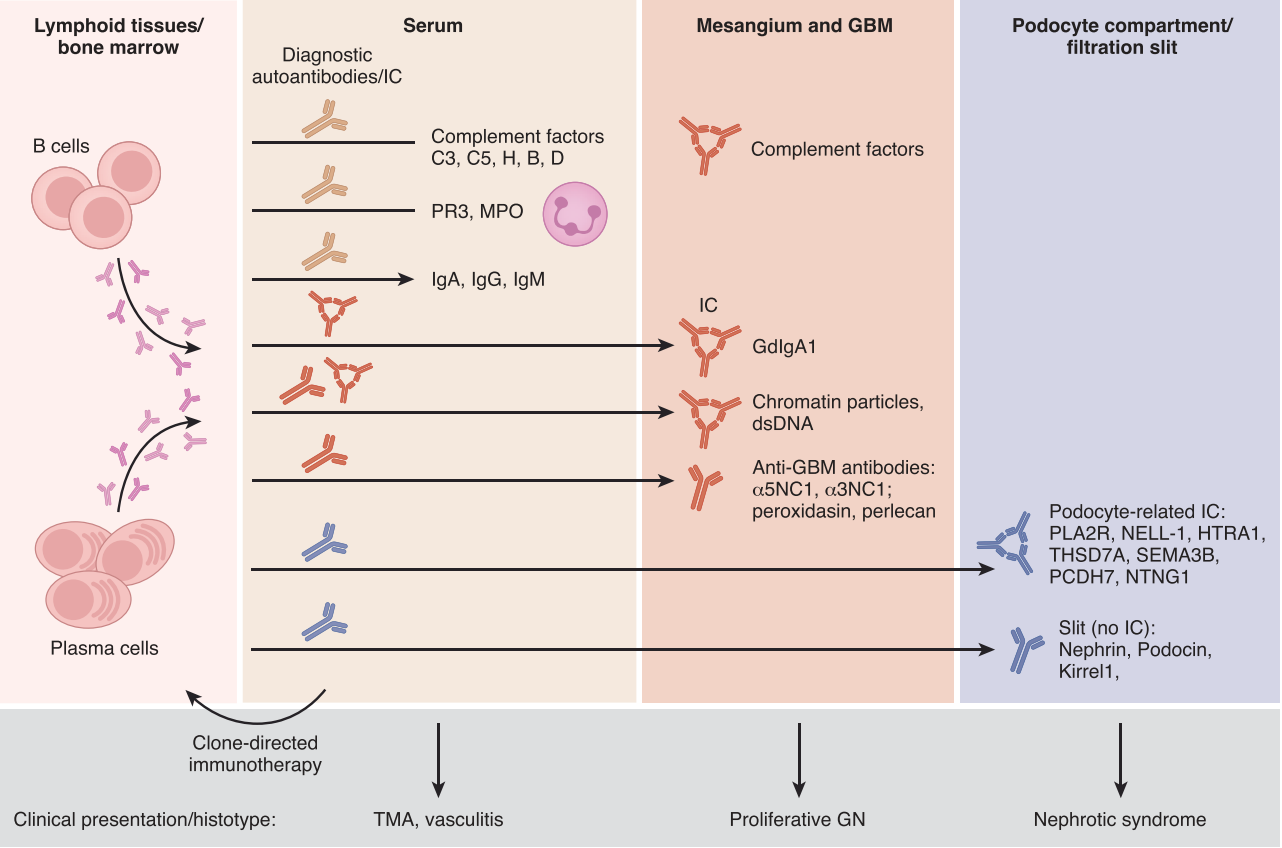

Figure 1. The spectrum of autoimmune GN and podocytopathies. Autoimmune GN is based on autoantigens, and as these are in different compartments, histotypes and clinical presentations differ, while the key elements of autoimmunity producing these autoantibodies, *i.e.*, B cells and plasma cells, are identical. Autoantigens inside the blood such as complement regulators, immunoglobulins, or neutrophil proteins trigger TMA or small vessel vasculitis. Antigens present in the mesangium or circulating IC that get trapped there cause proliferative GN as do antibodies binding to proteins inside the GBM. Antibodies against filtration slit antigens do not cause IC but cause nephrotic syndrome directly *via* signaling events, while antibodies against podocyte antigens form local IC and hence produce a membranous nephropathy histotype with podocyte injury causing nephrotic syndrome. No matter what histotype or clinical presentation, the nature of autoimmunity, and the involved elements of the innate and adaptive immune system are always the same; hence, treatment of autoimmune GN/podocytopathies is similar. dsDNA, double-stranded DNA; GBM, glomerular basement membrane; GdIgA1, galactose-deficient Ig A1; HTRA1, high-temperature requirement A serine peptidase 1; IC, immune complexes; MPO, myeloperoxidase; NELL-1, neural epidermal growth factor–like 1 protein; NTNG1, netrin G1; PCDH7, protocadherin 7; PLA2R, M-type phospholipase A2 receptor; PR3, proteinase 3; SEMA3B, semaphorin 3B; THSD7A, thrombospondin type 1 domain–containing 7A; TMA, thrombotic microangiopathy.

nature of the circulating nephrotoxic agent, usually an autoantibody (Figure 1). High-affinity, complement-activating IgG antibodies binding to critical structures inside the glomerulus can cause severe injury, inflammation, and rapidly progressive GN (RPGN). For example, anti-GBM antibodies induce severe kidney injury already at low serum titers.[20] Conceptually similar, antinephrin antibodies trigger nephrotic syndrome at low serum levels because they bind to critical signaling elements of the filtration slit.[11,21] In SLE, chromatin antibodies may be more or less nephropathic, leading to lupus nephritis that sometimes presents as mild focal mesangial proliferation or with severe crescentic lesions and focal tuft necrosis.[22] Antineutrophil cytoplasmic antibodies (ANCA) can be pathogenic even if not binding to glomerular structures.[23] ANCA nephrotoxicity depends on factors that relocate cytoplasmic proteins of neutrophils to the cell surface that eventually vascular injury through the formation of neutrophil extracellular traps and histone release. Thus, serum titers of ANCA correlate only loosely with the systemic activity of ANCA-associated vasculitis.[24] Infectious triggers also contribute. For example, anticomplement factor B antibodies produced during bacterial infections can cause acute postinfectious GN through concurrent complement activation and dysregulation.[25] When complement antibodies are present without a cotrigger, they may cause C3 GN (or atypical hemolytic uremic syndrome) only on additional complement system activation.[26]

Transient versus Persistent Autoimmunity

Transient autoimmunity is common during immune responses to pathogens[27] and remains mostly asymptomatic, although it can occasionally impair host defense as seen with anti-interferon antibodies in severe acute respiratory syndrome coronavirus 2 infection.[28] Postinfectious transient autoimmunity can manifest as symptomatic disease

in any organ system, for example, in postviral Guillain-Barre syndrome, acute disseminated encephalomyelitis, or myocarditis.[29–31] Among symptomatic glomerular diseases, several forms of transient autoimmunity with autoantibodies against either slit membrane proteins such as nephrin (steroid-sensitive nephrotic syndrome)[32] or podocytes (spontaneous remission of membranous nephropathy)[33] or the GBM (anti-GBM disease)[20] are known. Notably, these diseases do not relapse when immune memory fails to establish, *i.e.*, in the absence of antigen-specific memory B cells, memory T cells, or long-lived plasma cells residing in lymphoid organs and the bone marrow.[34] Treatment totally depends on the aforementioned nephrotoxicity of the transient autoantibody: It is intense only in case of anti-GBM,[20] moderate in nephrotic syndrome due to anti-slit antibodies (no immune complexes),[8,35] or may even only be supportive in nephrotic syndrome caused by podocyte antigens when immune complexes form and a transient membranous lesion pattern establishes.[36] The distinction between transient versus persistent autoimmunity is essential to decide whether or not maintenance immunosuppression is needed. Where available, serum assays detecting and quantifying circulating nephrotoxic agents (*e.g.*, in PLA2R-related nephrotic syndrome) have transformed disease monitoring and therapy stratification.[37] Therefore, development of new assays where they are currently not commercially available (*e.g.*, in IgA nephropathy and anti-slit protein nephrotic syndrome) represents a critical research priority.[10,11,21,32]

In the next section, we will discuss that treatment approaches of autoimmune GNs should be guided primarily by disease activity rather than the type of antigen or histopathologic lesion pattern *per se*.[19]

Rapid Progressive Autoimmune GN

RPGN is defined as GN that leads to a rapid decline of excretory kidney function typically due to severe glomerular necroinflammation presenting with either vascular obliterations, loop necrosis, or crescents that obstruct the glomerular outflow.[38,39] Proteinuria levels vary, since podocyte injury is often a secondary event and thus considered a second-line parameter when assessing treatment responses.[38] Treatment goals in RPGN are three-fold (Figure 2).

Stopping Glomerular Necroinflammation Immediately

Necroinflammation in RPGN is primarily driven by intraglomerular complement activation,[39] which is not (effectively) suppressed by glucocorticoids or immunosuppressive drugs such as cyclophosphamide, mycophenolate, or even by plasma exchange.[40] As a result, treatment of RPGN usually results in some degree of irreversible kidney injury[41] due to ongoing necroinflammation despite intense immunosuppressive therapy. The traditional approach attempts to control the disease process only through blocking the *de novo* production of autoantibodies but does not immediately terminate glomerular necroinflammation.[23] Proof-of-concept data demonstrate that effective C5 inhibition can rapidly resolve hematuria, leukocyturia, and kidney dysfunction in RPGN.[42,43] Mechanistically, complement inhibition can shut down glomerular necroinflammation within hours, critical in RPGN due to high titers of very nephrotoxic autoantibodies. The success of complement inhibitors has been proven to be effective in chronic GNs[44,45] and underlines the urgency to test and implement such therapies in RPGN to minimize irreversible kidney injury in an unprecedented manner.

Secondary Glomerular Inflammation

Complement activation generates pro-inflammatory fragments such as C3a and C5a, which act as chemoatractic factors and promote leukocyte recruitment into the glomerulus, local cytokine and chemokine release, and neutrophil extracellular trap formation.[46,47] Clinical data obtained with avacopan added to the standard-of-care therapy in ANCA vasculitis demonstrated that blocking C5aR can improve kidney outcomes in autoimmune RPGN.[48] It is of note that avacopan is not a complement inhibitor *per se* because it leaves the C3a–C3aR axis and the cytotoxic C5b-9 complex intact.[44,49] Thus, avacopan may serve as a safer alternative to corticosteroids, which are associated with significant adverse effects.[48] It is tempting to speculate that real complement inhibitors may eventually replace both avacopan and corticosteroids as more potent agents to immediately stop necroinflammation and secondary inflammation.[46] Currently, corticosteroids are still widely used despite their toxicity, highlighting the need for alternatives such as avacopan and true complement inhibitors to improve efficacy and safety of early RPGN therapy.

Fast Removal of Circulating Highly Nephrotoxic Autoantibodies

Immediate removal of circulating nephrotoxic antibodies is advantageous in RPGN such as anti-GBM disease or cryglobulinemic vasculitis but remains controversial in ANCA-related RPGN and is not recommended in lupus nephritis.[50,51] Traditionally, plasma exchange or immunoabsorption has been used to achieve this aim, but these procedures require central catheter access and carry risks for infections, bleeding, or thromboembolic complications. Imlifidase, a bacteria-derived degrading human IgG-degrading enzyme, offers a novel alternative to achieve rapid depletion of circulating IgG, including nephrotoxic immunoglobulins.[52] While promising, repeated administration is limited by potential allergic reactions and high costs.[53]

Preventing Relapse in Cases of Persistent Autoimmunity

When RPGN is associated with the formation of clones from memory B/T cells and long-lived plasma cells, sufficient doses of a maintenance immunosuppression is required, conceptually like managing persistent alloimmunity in kidney transplantation.[23] The principles of such maintenance immunosuppression are always similar and are discussed in the next section. Other forms of RPGN, for example, anti-GBM disease or postinfectious RPGN, are usually associated with transient autoimmunity and do not require long-term immunosuppression because memory B/T cell clones and long-lived plasma cells do not form.[20]

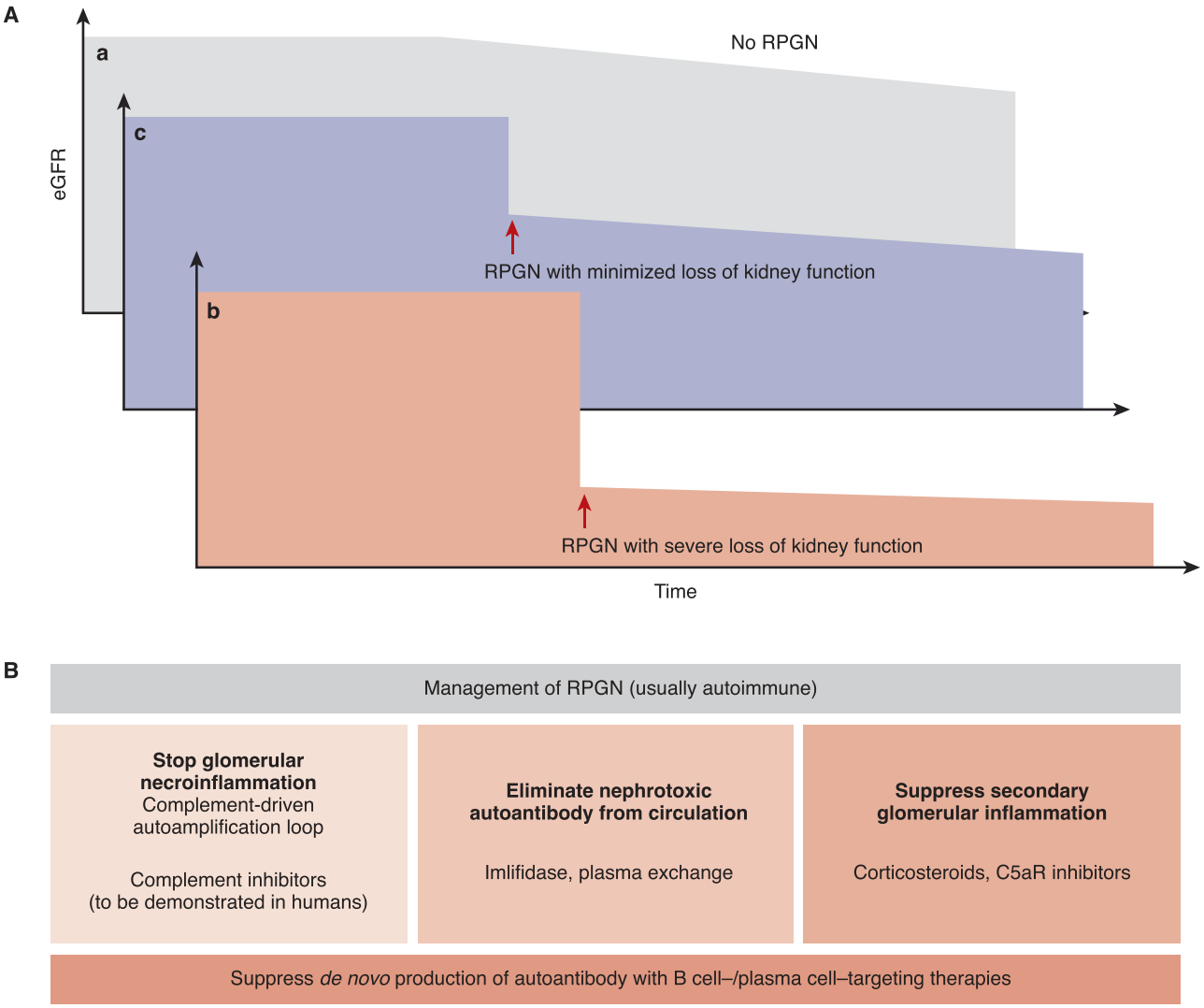

Figure 2. RPGN. (A) In RPGN, time matters to minimize irreversible glomerular injury. (B) On day 1, stop glomerular necroinflammation, eliminate the nephrotoxic autoantibody from the circulation, and suppress secondary glomerular inflammation. Day 2 is still early enough to suppress *de novo* production of the autoantibody in the lymphoid tissues with immunosuppressants or specific B-cell– and plasma cell–targeting therapies. RPGN, rapidly progressive GN.

Altogether, RPGN is a serious disease associated with irreversible kidney injury, that is, CKD or kidney failure as well as serious therapy-related morbidity and mortality.[38] However, the therapeutic landscape is evolving rapidly. The integration of complement inhibitors and imlifidase may soon enable clinicians to replace or minimize the adverse effects of corticosteroids and plasma exchange as well as irreversible kidney injury (Figure 2).

Relapsing Autoimmune GN or Podocytopathies

Relapsing autoimmune GN or podocytopathies are manifestations of persistent autoimmunity.[1] This fact often conflicts with the desire of patients and doctors to taper or discontinue therapy following initial or recurrent disease relapse. Consequently, many relapses are, in part, self-inflicted, not embracing the mechanisms of immune memory requiring long-term immunosuppression. However, persistent autoimmunity is conceptually similar to persistent alloimmunity after kidney transplantation,[1] where no one questions the concept of life-long immunosuppressive therapy to prevent graft rejection episodes.[54] Just as the irreversible injury inflicted by 2–4 episodes of rejection episodes exhaust function of the allograft, the same applies also to repeated relapses in patients with autoimmune GN.[55] Three treatment goals can be distinguished (Figure 3).

Induction of a First Remission

Initial therapy of relapsing autoimmune GN follows evidence-based treatment standards.[18] If a single immunosuppressant or a dual or triple combination is needed depends on the number of autoantigens and hence the

autoreactive clones that need to be controlled.[56] For example, antinephrin or anti–PLA2R-induced nephrotic syndrome generally respond well to monotherapy with rituximab because only single clones of autoreactive lymphocyte clones must be controlled.[32,36] By contrast, patients with lupus nephritis, where hundreds of autoantigens (=autoreactive lymphocyte clones) are present, benefit from combination therapy at onset, *e.g.*, corticosteroids, mycophenolate, belimumab, voclosporin, or obinutuzumab.[51] Very active or severe autoimmune GN/podocytopathies that present either with very severe nephrotic syndrome and high autoantibody titers or as RPGN require additional means, as discussed before.[57] The main treatment challenge in relapsing forms of autoimmune GNs is to prevent disease relapse as each flare reduces kidney lifespan and is associated with a higher risk for cardiovascular disease, kidney failure, and early mortality.[58] This shift in focus already adopted in ANCA-associated vasculitis should guide future management strategies in all relapsing GNs.[59]

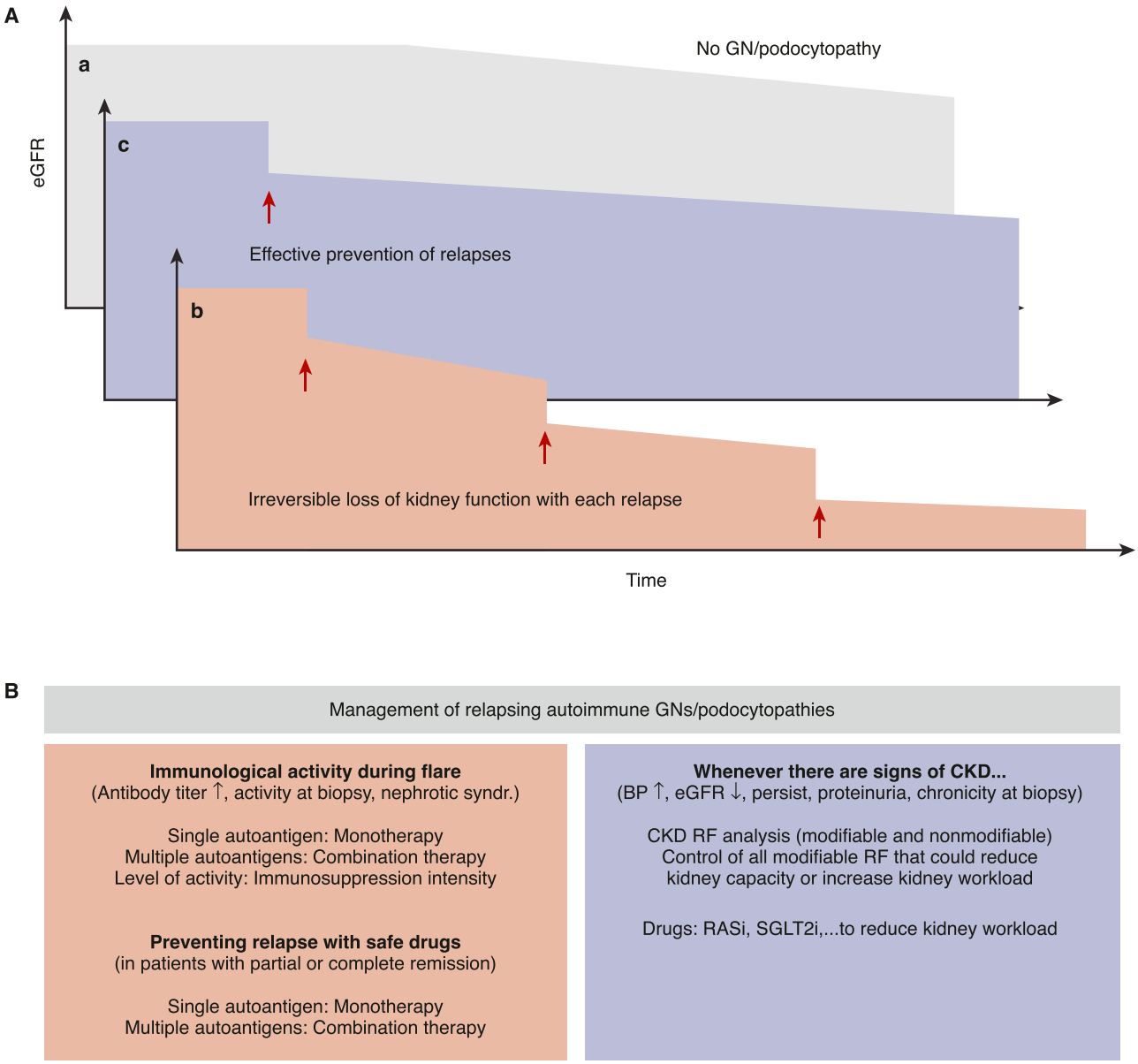

Figure 3. Relapsing GN/podocytopathies. (A) The main concept is divided into controlling activity with immunotherapy and treating CKD with the usual CKD interventions. (B) Controlling activity occurs in two phases. During an active episode, the level of activity and the number of autoantigens (number of autoreactive lymphocyte clones to control) determine the choice of immunotherapy. As relapsing disease is a sign of persistent autoimmunity, usually long-term immunosuppression is needed even in patients with complete clinical remission to prevent further relapses. Combination therapy is more efficient in patients with multiple autoantigens. Toxic drugs should be replaced by safe drugs in this phase. RASi, inhibitors of the renin-angiotensin-aldosterone system; RF, risk factor; SGLT2, sodium-glucose transporter 2.

Persistent Suppression of Memory Clones to Prevent the Next Flare

Relapsing autoimmune GN or nephrotic syndrome is a clear sign of chronic autoimmunity characterized by the presence of autoreactive memory B/T cells and long-lived plasma cells that require a better immunosuppression.[1] In diseases such as SLE, autoimmunity is inherently chronic and the formation of autoreactive memory clones is expected. Therefore, an adequate dose of maintenance immunosuppression targeting these clones should be initiated already upon initial diagnosis.[60] In lupus nephritis, an important treatment goal is to prevent a relapse.[58,60] If it does, exposure to therapy was insufficient either due to inappropriate dose tapering or patient drug nonadherence.[61] It is therefore useful to follow a treatment approach similar to the kidney transplantation protocol, *i.e.*, using a long-term combination therapy, given the high number of autoreactive lymphocyte clones in SLE.[56] This contrasts the "simple" relapsing autoimmune glomerulopathies with single autoantigens (*e.g.*, proteinase 3, myeloperoxidase, PLA2R, nephrin).[56] A main reason for the tendency to taper therapy despite persistent immune memory is the fear of drug toxicity as unselective immunosuppressants, namely, corticosteroids and mycophenolate mofetil, are still in use.[62] Such toxic drugs mandate dose-tapering, implying phases of undertreatment and hence disease relapses. The increasing availability of drugs with an excellent safety profile, such as hydroxychloroquine and belimumab, have been shown to prevent disease relapses even at long-term use.[63–65] Similarly, low-dose rituximab is effective in the maintenance phase of ANCA vasculitis.[66] However, hypogammaglobulinemia remains a concern in these patients, and it is unlikely to improve even with deeper B-cell depleters such as obinutuzumab.[67] B-cell deactivators, such as belimumab or other future nondepleting B-cell–targeting agents, may help to solve this problem.[68] Patients with frequent relapsing nephrotic syndrome face the same problem and may benefit from deeper B-cell depletion or from targeting plasma cells as a source of the nephropathic autoantibodies.[69]

Treating CKD in Relapsing Disease

Patients with relapsing GN should be regarded as patients with CKD from the very first episode.[70] Each flare leads to cumulative, irreversible kidney injury, increasing the workload to the remaining nephrons and accelerating disease progression.[70] Patients, for example, with lupus nephritis managed by a rheumatologist must also receive dedicated CKD care by a nephrologist to assure that the patient benefits from the latest developments in CKD management.[70] The same applies to all patients with podocytopathies, especially when chronicity (*i.e.*, FSGS lesions) is evident.[71] Thus, a comprehensive assessment of CKD risk factors (RF) is critical.[70] For instance, prematurity at birth is an indicator of low nephron endowment, similar to a history of previous GN episodes or kidney injuries, impaired GFR, or interstitial fibrosis and tubular atrophy at kidney biopsy. All these are indicative of a low nephron number (kidney capacity), which explains higher levels of proteinuria due to compensatory hyperfiltration of the remaining nephrons.[72] Advanced age and male sex are additional RFs associated with a lower residual nephron number.[73] The other side of the CKD equation is the kidney workload. The dysbalance between kidney capacity and workload determines the slope of future GFR decline independent of further disease relapses.[72] With each GN relapse, this workload of the remaining nephrons increases further, making relapse prevention a primary therapeutic goal in relapsing glomerulopathies.[71] Concomitant or steroid-induced overweight or weight gain as well as high salt or protein intake increase this discrepancy between kidney workload and capacity as does hyperglycemia (diabetes).[72] Especially since the urge for Hba1c control in individuals with type 2 diabetes is no longer as tight as it used to be,[74] even guideline-treated patients may expose their remaining nephrons to additional hyperfiltration and metabolic overload.[75] Pregnancy represents another state of severe hyperfiltration.[76] Although this is well tolerated in young women with adequate nephron reserve, it can be detrimental in those with lower nephron capacity. In such cases, further GFR increases may not be accommodated, leading to (*1*) preeclampsia at some stage during pregnancy[77] and (*2*) a pregnancy-induced adaptive podocytopathy causing additional irreversible kidney injury.[9]

Thus, numerous lifestyle interventions are needed in patients with relapsing GN that are identical to what is considered routine CKD management.[71] CKD management includes drug interventions that reduce remnant nephron workload. Inhibitors of the renin-angiotensin system reduce hyperfiltration by dilating the efferent glomerular arteriole, a vasoeffective effect that reduces filtration fraction in the glomerulus.[71] Sodium-glucose transporter 2 (SGLT2) inhibitors, by blocking sodium and glucose uptake in the proximal tubule, directly decrease the metabolic workload of these cells.[78] In addition, SGLT2i reactivate the tubuloglomerular feedback mechanism, which reduces glomerular hyperfiltration through vasoconstriction of the afferent glomerular arteriole.[79] Consequently, dual therapy of renin-angiotensin system/SGLT2 inhibitors exerts synergistic effects on GFR decline in patients with relapsing GN, even when immune control is adequate.[80]

Taken together, chronic autoimmunity implies the risk of disease relapse. Since each flare results in irreversible nephron loss and accelerates progression to kidney failure, a focus on preventive immunotherapy is essential. The number of involved autoantigens (clones) and thus autoreactive lymphocyte clones should guide the level of immunosuppression. Importantly, all patients with relapsing GN/podocytopathies should receive CKD care.

Chronic Autoimmune GN/Podocytopathies

IgA nephropathy is the most common form of chronic autoimmune GN; however, other diseases such as autoimmune C3G or smoldering lupus nephritis, ANCA-associated vasculitis, and membranous nephropathy also fall within this category. These disorders are characterized by low-grade but persistent disease activity. Even in the absence of nephropathic autoantibodies levels, ongoing subclinical inflammation can drive chronic lesions resulting in gradual kidney damage, that is, nephron loss, interstitial

fibrosis, and progression to kidney failure.[81] Hence, two major treatment goals must be distinguished (Figure 4).

Treatment of CKD

All patients require optimal CKD care including rigorous control of modifiable CKD RFs such as education on salt and dietary protein intake, optimizing body weight, and diabetes management where applicable.[71,72] Any exposure to potential nephrotoxins needs to be avoided. Optimal CKD care implies the latest combination therapy using nephroprotective drugs to reduce the consequences of remnant nephron hyperfiltration.[82,83] Currently, dual renin-angiotensin system/SGLT2 inhibition has become the standard of care. Additional endothelin blockade with agents such as sparsentan has shown promise in IgA nephropathy.[84] Soon, the mineralocorticoid receptor antagonist finerenone, glucagon-like peptide 1 receptor agonists, and other novel drug classes may provide additional treatment options to minimize nonimmune factors driving the progression of CKD in these patients.[85] It is important not to miss the CKD aspect in the GNs, especially when eGFR is still in the normal range. Cystatin C, persistent proteinuria, and chronic changes at kidney biopsy inform about the presence of CKD in the GNs.

Control of Persistent Autoimmune Activity

The underlying immunologic activity originates from autoreactive lymphocyte clones located in various lymphoid tissue compartments. In IgA nephropathy, a distinct feature is the production of GdIgA1 in mucosal lymphoid tissues such

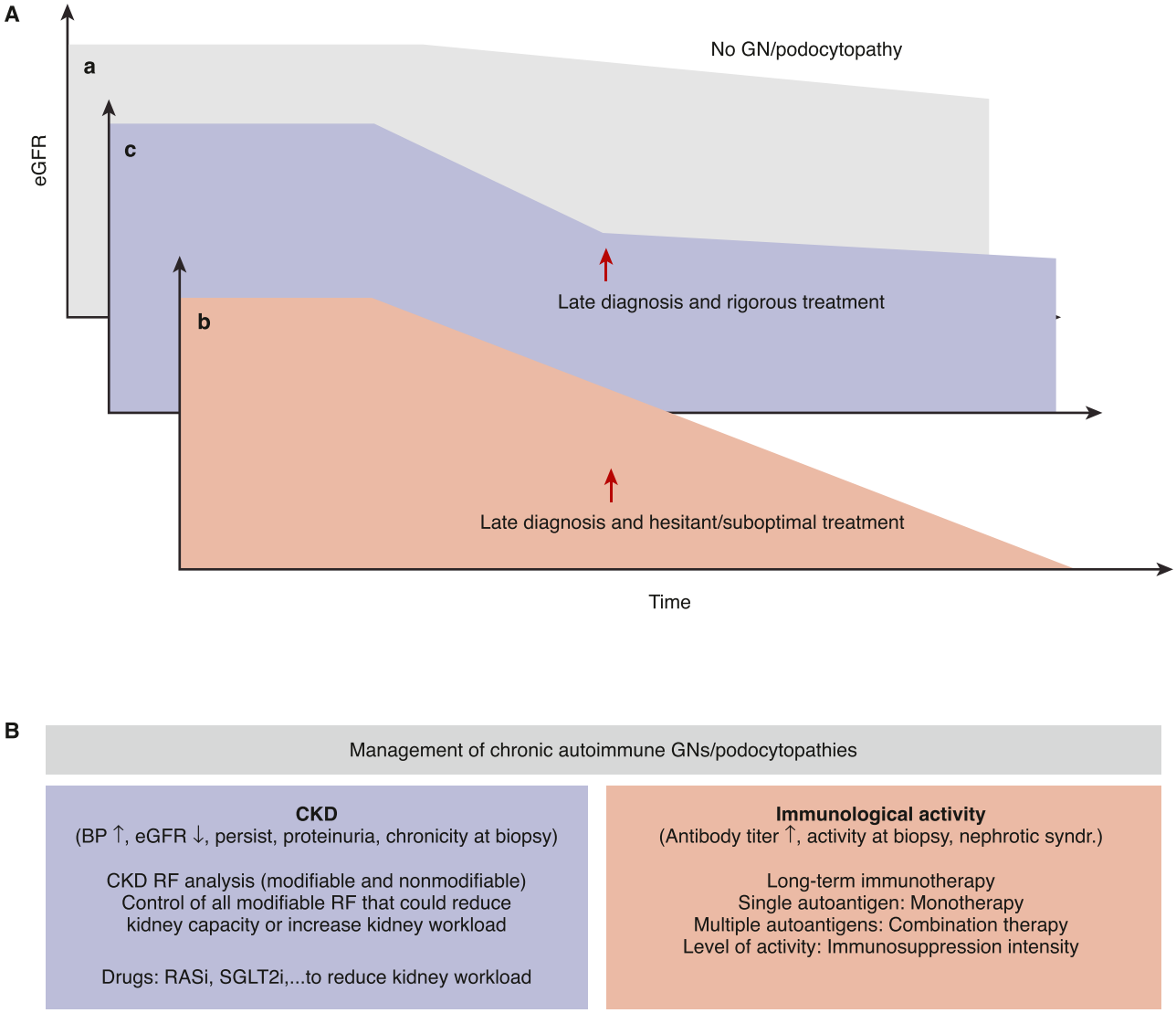

Figure 4. Management of chronic autoimmune GN or podocytopathies. (A) The therapy of CKD is the primary strategy in chronic GN or podocytopathies. (B) Signs of chronicity are elevated BP, a reduced eGFR, persistent proteinuria, or irreversible lesions at kidney biopsy, including FSGS. CKD care includes an individual RF analysis and control of all modifiable RF. Treatment goal is to minimize any further reduction of kidney capacity and a reduction of the workload of the remaining nephron with lifestyle interventions or drugs such as inhibitors of the RAS or SGLT2. Immunologic therapy is always long term and varies depending on the number of autoantigens and the level of activity.

as the tonsils and Peyer's plaques in the intestinal tract.[86] This provides a rationale for selectively targeting B and plasma cells of the mucosal immunity to minimize the production of the autoantigen, thereby attenuating all subsequent nephropathic events in IgA nephropathy and slowing kidney damage.[87] That is why drugs targeting B-cell activating factor or a proliferation-inducing ligand are particularly promising because these factors control both the antigen-producing cells as well as the autoantibody-producing cells, a unique constellation in IgA nephropathy among all autoimmune disorders. By contrast, therapies aimed at targeting only the autoantibodies against GdIgA1 have thus far proven less effective in attenuating IgA nephropathy.[88] In all other autoimmune GNs/podocytopathies, however, targeting cells that produce circulating autoantibodies is effective as the antibodies are either directly or indirectly nephropathic.[69] Standard immunosuppressive drugs remain central to therapy, with drugs choice guided by both efficacy and safety data from clinical trials.[7] Given the chronic nature of these diseases, long-term immunotherapy in chronic GNs is typically required, which limits the use of drugs associated with significant toxicities such as corticosteroids (metabolic and bone effects), calcineurin inhibitors (hypertension, nephrotoxic, and metabolic effects), mycophenolate (tumors), intestinal release budesonide (metabolic effects), and rituximab (hypogammaglobulinemia). These considerations rather favor drugs with fortunate safety profiles such as hydroxychloroquine and belimumab, but convincing trial evidence supporting their long-term use for many types of chronic GNs remains limited or lacking.

Summary

The traditional classification of glomerular diseases by histotypes has come to age because it fails to connect with the expanding treatment landscape and the shift toward personalized medicine. Many diseases are no longer treated based solely on nonspecific histologic lesion patterns but rather according to their underlying pathogenesis. The ongoing discovery of specific autoantibodies as biomarkers assuring the diagnosis, predicting prognosis, and defining how to use immunotherapies is of utmost important for this domain and requires broad research activities. Indeed, among the GNs, only the autoimmune GNs/podocytopathies consistently require immunosuppressive therapy. Even the conventional distinction of GNs and podocytopathies is becoming arbitrary; for example, membranous nephropathy represents both an inflammatory immune complex–mediated GN and a podocytopathy, underscoring the overlap in pathogenic mechanisms and the need for a more integrated conceptual framework as suggest herein. Finally, nephrologists providing GN care require advanced training on the elements and regulators of innate and adaptive immunity to keep up with the expanding treatment landscape of specific immunomodulatory drugs.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at http://links.lww.com/JSN/F482.

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Funding

H.-J. Anders: BMBF (Per-NEPH, 01KU2204) and Deutsche Forschungsgemeinschaft TRR332, project A7, AN372/29-1 and AN372/32-1 (no 554613516), STE2437/4-1 and STE2437/4-2. S. Steiger: Deutsche Forschungsgemeinschaft TRR332, project A7, AN372/29-1 and AN372/32-1 (no 554613516), STE2437/4-1 and STE2437/4-2. P. Romagnani: European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (grant agreement no. 101019891).

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